the same way, it's likely that a marketing strategy would work (or fail) consistently across the group (if segmentation is done properly).
An algorithm won't output labels for the groups, it won't tell us if a group is "students under 25" and another "mid-aged book writer, with a passion for comics": it will only gather similar people together, and then data analysts can further examine the groups to understand more about their composition.

2. **Finding outliers**: find data that stand out. Depending on the context, it could be noise in a signal, it could be a new species of flowers in a rain forest, or even a new pattern or behavior in customer analytics.
3. **Preprocessing**: find clusters in the data and process each cluster separately (and possibly in parallel); this obviously provides a speedup but it also has another side-effect: since it reduces the max amount of space needed at any single time, when a huge dataset is broken up into smaller pieces, each piece can fit in memory or be processed on a single machine, while the whole dataset couldn't. Sometimes you can even use a fast clustering algorithm (for instance *canopy clustering*) as a preprocessor step for a slower clustering algorithm.

12.1.2 Types of Clustering

Clustering is an NP-Hard[3] problem, and as such it is computationally difficult (impossible for today's real datasets) to solve it exactly; moreover, it is hard even to define objective metrics that can assess the quality of a